

Whose Regulation Does the Model Know? Reliability of LLM-Retrieved Regulatory Information for Public Environmental Management in 25 Countries

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Abstract

A municipal environmental manager in São Paulo, Lagos, or Jakarta asks a language model what the binding PM_{2.5} standard is in their country. Nothing in the answer says whether it matches the official register, and no established method exists for checking. We contribute such a protocol and report what it finds. It scores answers against official registers, adjudicating by code wherever the correct answer is a published value and by a three-vendor judge panel elsewhere. We apply it to air pollution regulation across 25 countries, 14 deployment stacks and four languages (8,244 scored cells). The three fixes an agency can deploy without building anything all fail, and the most intuitive one makes the answer worse: querying in the country’s own language *lowers* accuracy by 5.0 percentage points (Wilcoxon $p = 5 \times 10^{-16}$) and makes the model 9.4 points less likely to return a register-checkable value

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at all; declaring the official role does not narrow the gap ($p = 0.29$); and a regionally tuned model ranks last of fourteen. Accuracy also collapses on the task that carries administrative consequence, retrieving the binding value: 0.39 against 0.61 for synthesis and recommendation. A North/South gap persists (+5.0 pp, permutation $p = 0.021$), concentrated where answers depend on a national fact — on the national standard, scored by code against the instrument in force, Global South odds are 0.33 of Global North odds. Scope is bounded: we measure agreement with published regulation in one domain, not general capability. The protocol transfers to any regulated domain whose correct answer is published.

Keywords: algorithm auditing, geographic bias, multilingual bias, digital government, Global South

1. Introduction

Large language models have moved from research curiosity to working infrastructure of public administration in under five years: an OECD survey of core government functions documents generative-AI adoption across policymaking and service delivery, including assistants that help public-service advisers retrieve information and the sources behind it (OECD, 2025). Air pollution policy is a domain where this carries direct stakes for public health. Fine particulate matter is among the leading environmental risk factors for premature mortality worldwide (GBD 2019 Risk Factors Collaborators, 2020; World Health Organization, 2021), and the burden falls disproportionately on the Global South. When a public environmental manager in São Paulo, Lagos, or Jakarta asks an LLM about an ambient air quality standard, an

attainment deadline, or the evidence behind an intervention, an inaccurate answer is not a benign error. It can propagate into a regulatory brief, a procurement decision, or an emergency response during a pollution episode.

This paper asks whether LLMs serve Global South air pollution policy as reliably as they serve the Global North, and whether the way a query is framed changes the answer. Both premises are established. Geographic factual bias is real and measurable: error rates are $1.5\times$ higher for Sub-Saharan African than North American countries across 20 models (Moayeri et al., 2024), model ratings on sensitive subjective topics track socioeconomic conditions up to $\rho = 0.70$ (Manvi et al., 2024), and models systematically privilege statements about the Global North (Mirza et al., 2024). And the linguistic asymmetry behind it is structural: the languages of most Global South countries sit in the under-served classes of the resource hierarchy (Joshi et al., 2020), and large Internet-derived corpora “overrepresent hegemonic viewpoints and encode biases potentially damaging to marginalized populations” (Bender et al., 2021). The asymmetry reproduces the uneven epistemic infrastructure decolonial scholarship has analysed for AI systems (Mohamed et al., 2020).

Three gaps remain open for a domain like this one, and they organise the study. No benchmark targets regulatory information with official ground truth: existing evaluations use World Bank recall, subjective ratings, journalistic fact-checking, or everyday cultural knowledge, none of which measures the tasks a manager actually delegates — stating a binding standard, retrieving an officially measured concentration, summarising health-burden evidence, identifying legal instruments and executing agencies, recommending

an operational response. Persona framing is unexamined as an experimental lever, although practitioners routinely prefix a role and the vendors themselves advise it (Section 2); whether it *reduces* the gap by steering toward regulation-compliant answers or *amplifies* it by inviting confident fabrication has not been tested in a pre-specified design. And the corpus-representation mechanism is asserted rather than measured against confounders, and rarely separated into its two channels — *language-corpus* size (Xue et al., 2021; Abadji et al., 2022) against *country-coverage* breadth — a distinction that matters here because the geographic gap is measured in English.

Objectives and scope

The study has three bounded objectives. *First, to establish a protocol for auditing regulatory information:* the object is agreement between a model’s answer and the value published in the official register, for tasks an environmental agency actually delegates — not model capability in general, reasoning quality, or usefulness for any other purpose. The protocol is the contribution that transfers; the domain is where we demonstrate it. *Second, to test the mitigations available to an administration:* a public body facing poor answers about its own country can realistically declare the official role in the prompt, ask in the national language, or procure a regionally tuned model, and we test all three as designed factors rather than discussing them. We do not test retrieval augmentation, fine-tuning, or human-in-the-loop verification, which require capacity most agencies in our Global South sample do not have. *Third, to locate where reliability differs and, as far as the design allows, why:* we test the corpus-representation account at the level where the measurement has resolution, and claim no causal identification.

Throughout, the unit of analysis is the model *as served* — weights, vendor, and serving infrastructure reached through a pinned endpoint — because a public manager interacts with that stack and any measured gap is a property of it. Hypotheses, sampling and analysis plan were fixed before data collection (Nosek et al., 2018) for 15 countries and then extended to 25; the plan was never deposited publicly, so we do not claim pre-registration, and the study is reported as exploratory throughout.

Contributions of this work

1. *An auditing protocol for regulatory information, not a trivia benchmark.* The question an administration faces is not whether a model writes well about policy but whether the value it returns matches the official register. We build five task types from the everyday work of an environmental agency (Table 1), each keyed to an official source, so correctness is adjudicable rather than a matter of expert impression, and the protocol transfers to any regulated domain where the correct answer is published.
2. *The three localisation fixes agencies reach for, tested rather than assumed — and all three fail.* Persona framing, native-language prompting and a regionally tuned model are each a procurement or workflow decision, and each circulates as received wisdom. None works, and the one a manager is most likely to try first does active harm: asking in the country’s own language lowers accuracy by 5.0 percentage points, most sharply for the language with the scarcest corpus.
3. *A design that separates who is poorly served from why.* Countries are stratified on independent axes — the UNCTAD developed/developing

split, the Joshi linguistic-resource taxonomy, and development — so geographic, linguistic and economic explanations can be told apart. Between countries, coverage and development are collinear and cannot be separated; within countries, the task-level contrast rules out the language-corpus channel but still cannot choose between coverage and development, and we say so. We also flag that the pre-specified corpus proxy is defective in a way worth knowing for anyone who has used it: it assigns the English Wikipedia to every sample country with English as an official language, indexing colonial history rather than corpus size.

4. *Ground truth adjudicated by code, and what that changes.* Rebuilding two task keys from official registers and moving range comparison into code cut the development gradient from $\rho = 0.51$ to $\rho = 0.36$ and more than doubled the native-language penalty, on the same responses; moving the third register-valued task from a judge to code removed a confound between judge and tier that the judged scores had carried. In this literature the ground-truth key, not the sample size, is the binding constraint.
5. *Open resources.* Protocol, prompts, ground-truth registries with their official sources, analysis code, and the deviation log are released under CC-BY-4.0 and MIT, with a one-command reproduction pipeline.

2. Related Work

Prior work runs along four threads: the measurement of geographic factual bias, multilingual and cultural extensions, mitigation attempts, and LLM use in policy and health decision contexts.

2.1. Geographic factual bias and multilingual extensions

[Manvi et al. \(2024\)](#) established that frontier LLMs carry systematic biases against regions of lower socioeconomic conditions, with correlations up to 0.70 between model-generated ratings and a proxy for those conditions — though the outcome space is subjective ratings, which leaves open whether the same bias appears in objective recall of regulatory facts. [Moayeri et al. \(2024\)](#) quantified factual recall against World Bank indicators across 20 LLMs, documenting $1.5\times$ higher absolute relative error for Sub-Saharan Africa than North America; its design is bounded by the macro-economic indicator space, while an environmental manager asks narrower, source-specific questions whose ground truth lives in environmental councils and monitoring agencies rather than one aggregated database. [Mirza et al. \(2024\)](#) showed that newer GPT-4 versions do not monotonically improve and that the privileging of Global North statements persists.

On the multilingual side, [Myung et al. \(2024\)](#) covered 16 countries and 13 languages in everyday cultural knowledge, finding that LLMs perform better in the native languages of high-resource cultures and worse in those of low-resource ones — the finding that motivates H2 and connects to the resource hierarchy of [Joshi et al. \(2020\)](#). Regional benchmarks have proliferated: BRoverbs ([Almeida et al., 2025b](#)), TiEBE ([Almeida et al., 2025c](#)), ALBA ([Vieira et al., 2026](#)) and AMALIA ([Simplício et al., 2026](#)) document concrete failures in culturally grounded tasks but do not address applied policy use. Work on Portuguese corpus construction is explicit that the effort has centred on English and that building effective corpora for other languages remains open: [Almeida et al. \(2025a\)](#) assemble a 120B-token Portuguese corpus

precisely because language-specific pipelines are needed to reach industrial quality. That asymmetry in how much text exists per language is what H4 tests.

These studies establish that the phenomenon exists and is measurable. None isolates a single applied domain with verifiable official ground truth, and none treats prompt framing as a manipulable factor.

2.2. Mitigation, persona, and role conditioning

[Opuszko and Böhm \(2026\)](#) found that reasoning helps in aggregate but does not eliminate regional disparities, and [Kerche et al. \(2026\)](#) proposed a place-based typology situating geographic bias in a broader sociotechnical landscape. Both are conceptual and aggregate.

Role conditioning is not a folk practice but what the vendors instruct their own users to do: Anthropic documents role setting in the system prompt ([Anthropic, 2026](#)), OpenAI’s guide opens the recommended message structure with an identity block ([OpenAI, 2026](#)), Google’s Gemini documentation asks that role definitions — which it calls persona — be placed in the system instruction ([Google, 2026](#)), xAI’s standard example initialises every conversation with a role-assigning system prompt ([xAI, 2026](#)), and the pattern is catalogued in the prompting literature ([White et al., 2023](#)). The systematic evidence on whether it helps is sobering: across four LLM families and 2,410 factual questions, expert personas did not improve accuracy over a no-persona control and slightly *reduced* it across 162 roles ([Zheng et al., 2024](#)). An independent study with a different design — six models on graduate-level multiple-choice questions — reaches the same conclusion: matching an expert persona to the problem type had no significant effect on performance ([Basil](#)

[et al., 2025](#)). Whether persona conditioning helps or harms factual reliability in high-stakes domains is therefore unsettled and, to our knowledge, untested for geographic equity. We treat persona as a pre-specified experimental factor and test both the Norm-Compliance hypothesis (persona reduces the gap) and the Persona-Vacuum hypothesis (persona amplifies it).

2.3. LLMs in policy, health, and the administration

[Le Coz et al. \(2025\)](#) evaluated LLM policy recommendations for homelessness across four cities — Barcelona, Johannesburg, South Bend and Macau — and read the pattern as “a context-blind rigidity and a potential bias based on the published online discourse on homelessness in the regions of the world with overrepresented data in LLMs”, against experts who tailored decisions to locally encoded realities. That is one domain and one task type, which we generalise to five task types spanning recall, synthesis and recommendation. In health, [Kozlakidis et al. \(2026\)](#) argue that hallucinations carry acute risk in regulatory environments with limited technical capacity, where language and cultural mismatch increases error, a risk that meets the mortality burden documented by the Global Burden of Disease estimates ([GBD 2019 Risk Factors Collaborators, 2020](#)).

The deployment question is no longer hypothetical. [Kim \(2026\)](#) estimates bureaucratic productivity effects of generative AI quasi-experimentally, and [Robinson \(2026\)](#) documents that choosing which model an agency runs is now an explicit procurement decision between open-weight and vendor-hosted options — a decision both our model-level findings speak to, since the open-weight models as a group trail the closed ones by 12.9 points on the composite even though one open-weight model (DeepSeek-V3) leads the roster, and the

regionally tuned Portuguese model is the weakest of the fourteen. Two further strands frame why a wrong answer matters administratively: [Choroszewicz \(2026\)](#) shows that generative-AI support tools position frontline practitioners as “moral crumple zones”, absorbing responsibility for failures originating upstream, and [Cordella and Gualdi \(2024\)](#), analysing the first binding measure a regulator imposed on a generative-AI service — Italy’s intervention on ChatGPT — argue that technology-neutral frameworks overlook what is specific to these systems and that regulators must engage their technical workings directly. On reception, [Aoki et al. \(2024\)](#) test whether the type of explanation changes perceived accuracy, fairness and trustworthiness among those affected.

What none of this establishes is whether the underlying answer is *correct*, and whether correctness is distributed evenly across the states that rely on it. Perceived trustworthiness and measured accuracy are different quantities, and a system can be well explained, well governed, well received, and wrong. That is the gap this study addresses, in a domain where the correct answer is published in an official register and can be checked.

2.4. Positioning of the present work

Our design differs from prior work in four ways: (i) a single high-stakes applied domain with verifiable official ground truth, rather than numeric trivia or cultural knowledge; (ii) persona as a pre-specified within-subject factor rather than an uncontrolled framing choice; (iii) theory-driven country sampling on independent axes (UNCTAD coloniality, Joshi linguistic resource, development) rather than regional convenience; and (iv) an explicit corpus-representation mechanism test with sensitivity analysis. No prior study com-

bines a regulated public-health domain, a persona manipulation, and a pre-specified mechanism test.

3. Methods

3.1. Research design

We specify a pre-specified, factorial, multi-country benchmark experiment quantifying geographic and persona-modulated bias in 14 large language models on air pollution policy tasks. The pre-specified design crosses 15 countries (stratified on three theoretically grounded axes), 14 LLMs spanning five access tiers, one domain, 5 task types and 2 persona conditions, with 2 stochastic replications per cell; a post-hoc extension (Section 3.3) enlarges the sample to 25 countries. The stimulus set crosses each country’s five tasks and two personas in English, plus a within-country native-language rendering for the nine countries with a target native language, scored on the $[0, 1]$ composite. The protocol fixes, in advance of collection, the hypotheses, primary models, persona protocol, exclusion criteria and multiple-comparison corrections. A Brazil-only collection executed under an earlier three-domain design is reclassified as an extended pilot and reported separately; it informed prompt calibration but contributes no confirmatory observations.

3.2. Country selection

Countries were selected by theoretical sampling (Patton, 2015) triangulated across three independent classifications: the Global North/South distinction, operationalised through UNCTAD’s developed/developing economies classification (UNCTAD, 2024), which carries the coloniality-of-knowledge framework motivating the study (Mohamed et al., 2020); the

Joshi et al. six-class taxonomy of linguistic-resource representation (Joshi et al., 2020); and World Bank income groups (World Bank Group, 2025).

The pre-specified sample of 15 — Brazil, Mexico, Argentina, Peru; Nigeria, South Africa, Kenya, Egypt; India, Indonesia, Bangladesh, the Philippines; and the United States, Germany, Japan as Global North references — covers four world regions and four income groups. The linguistic-resource axis turned out far less differentiated than intended: the dominant official languages fall into only three Joshi classes, twelve of the fifteen sharing Class 5, and extending to 25 does not help since eighteen are Class 5. This is a property of the world rather than of our draw, because the official languages of most states are themselves high-resource, but it means the linguistic axis has too little variance to be separated from the others, and we treat the Joshi results accordingly (Section 4.6). Countries with compromised statistical governance or without a publicly retrievable national standard were excluded from the pre-specified fifteen; Oceania entered only with the extension (Australia), and Joshi Class 0 languages are not represented, which we note as a scope limitation.

3.3. Sample expansion after the pre-specified analysis

After the pre-specified 15-country analysis, and *reported here as a post-hoc extension rather than part of the locked protocol*, the sample was enlarged to 25 countries under identical procedures, to raise power for the gradient and mechanism tests and to multiply the within-country language pairs available for H2. The manuscript reports the 25-country sample throughout; the primary family recomputed on the pre-specified 15 alone is tabulated in Supplementary Material, so a reader can see what the locked protocol would

have concluded. Because the extension followed observation of the initial results, no pre-specified conclusion is revised on the extension alone, and the expansion is treated as a power-and-robustness check rather than a new confirmatory test. Angola, like Nigeria and Argentina, has *no* national ambient PM_{2.5} standard, adding no-standard cases in which a faithful model must decline to state a non-existent threshold. The ten additions and their roles are detailed in Supplementary Material. Two extra question variants per (country, task) were generated for the H2 pairs at design time but were never collected or scored; the H2 analysis uses the single native-language rendering of each English prompt.

3.4. *Country covariates*

Human Development Index (UNDP 2023–24) and GDP per capita index developmental position and enter the mechanism analysis as adjustment covariates. The corpus-representation exposures divide along a distinction the mechanism analysis makes explicit, because the two channels apply to different effects. *Language-corpus* measures quantify how much pretraining text exists *in* a language and are the candidate mechanism for the native-language penalty: mC4 token count (Xue et al., 2021, Table 6), OSCAR-2301 corpus size (corpus introduced by Abadji et al., 2022; per-language figures read from the OSCAR-2301 dataset card, whose totals differ from the 22.01 release described in that paper), and the Joshi class (Joshi et al., 2020). *Country-corpus* measures quantify how much the encyclopedic corpus is *about* a country, independent of language, and are the candidate mechanism for the geographic gap: the English Wikipedia article length, the number of Wikipedia language editions with an article on the country, and the

number of Wikidata statements on the country entity.

The two families are not interchangeable. Because the geographic gap is measured *in English*, with language held constant across countries, a residual North/South gap cannot be a language-corpus effect. One point deserves stating plainly, because the roles are easy to conflate: Wikipedia and Wikidata are never ground truth here, they are the *explanatory* variable. What an answer is scored against is always an official register (Section 3.9); had we scored against an encyclopedia, the study would measure agreement with an encyclopedia, which is not the question a public administration faces.

3.5. Model selection

Fourteen LLMs were benchmarked across five access tiers, spanning roughly two orders of magnitude in parameter count and four training-data geographies (United States, China, Europe, Brazil): open-weight frontier (5), open-weight mid (3) and open-weight small or regional (2), closed-accessible (3) and closed-frontier (1). Only three of the open-weight models were served locally (Ollama, pinned tags: Phi-4 14B, Qwen3 14B, Cabra-Mistral 7B); the other seven open-weight models were served through hosted endpoints (Groq free tier for five, the DeepSeek and Cohere APIs for two), and the four closed models through their vendors’ APIs. The small tier includes a Brazilian-Portuguese instruction-tuned model (Cabra-Mistral 7B v3), tested against a scale-matched globally trained open model for H3. Each model was queried with an identical pinned tag throughout the collection window. The full roster is in Supplementary Table S1.

We compare *models as served* through a fixed access stack, not idealised weights. Provider infrastructure, serving configuration and API layer are part

of what each label denotes, and a residual gap could in principle reflect the stack rather than the weights; we hold this constant by pinning version tags and a fixed sampling configuration per provider throughout a fixed window (Section 3.10).

3.6. Domain and task taxonomy

The single domain is air pollution policy, chosen because it anchors to officially published, verifiable ground truth and maps to concrete decision activities of a public environmental manager. The five tasks were not written as a generic knowledge quiz: each corresponds to a question a manager has to answer in the course of ordinary work, and each was drafted so that a wrong answer would carry an administrative consequence rather than merely lose a point. Table 1 gives them with the prompt exactly as issued to the models, using Brazil as the worked instance, alongside the decision each stands in for.

The tasks are ordered by how much of the answer is fixed by an official source. T1, T2 and T4 admit verifiable ground truth and carry the primary accuracy signal; T3 and T5 are rubric-scored open-generation tasks, where competent answers legitimately differ. That ordering is what makes the task dimension informative rather than decorative: it lets us ask whether models fail uniformly or fail specifically where a published national value has to be recalled, which is the question the results turn on. The country-by-task mapping of official source classes and verifiability status is in the ground-truth registry (Section 3.9), and the full 25-country prompt set is released with the data.

3.7. *Persona manipulation*

Persona is a within-prompt factor with two levels, and the manipulation is not arbitrary: assigning a role to the model is the practice the vendors themselves document and recommend (Section 2). The neutral condition presents the query as a plain information request; the `public_manager_env` condition prepends a fixed role frame identifying the requester as a municipal or regional environmental management official, holding the substantive question identical. For Brazil’s T1 the frame is the single sentence “*I am the municipal environmental management official of São Paulo, Brazil, and I need to brief decision-makers.*” A single template is applied uniformly across countries and tasks, translated where the prompt language is non-English, so any between-condition difference is attributable to the frame rather than to question wording. Persona order within each prompt-model cell is randomised, and a manipulation check verifies that the frame is acknowledged in the response. Exact wording, translation procedure and manipulation-check coding are in Supplementary Material.

One implementation detail matters for how the result should be read. Every request in this study is sent as a single user turn with no system prompt, so the role frame is prefixed to the user message rather than installed as a system instruction. Three of the four vendor guides place the role there; Google’s is explicit that it belongs *either* in the system instruction *or* “at the very beginning of the user prompt” (Google, 2026), which is exactly the channel we test. This is deliberate: the system-prompt route requires API access and a developer to configure it, while the public servants whose decisions motivate this study reach these models through the consumer chat

interface, where the only available channel is the message they type. We therefore test role framing as the sector actually performs it, and Section 5.7 states what remains untested. Persona is the basis of H6.

3.8. Prompt construction and validation

Prompts were authored for the domain by the research team with domain supervision and validated against the ground-truth registry before entering the confirmatory stimulus set, with every factual claim in a registry entry required to resolve to an official source URL or a peer-reviewed DOI before the corresponding prompt was admitted. The analysis plan called for a 50-prompt pilot scored by two independent human raters ($\alpha \geq 0.70$) before scaling the design. *That pilot was not run*, and no rater-level α exists for this study; rubric reliability is evidenced instead on the confirmatory data itself, by a three-vendor panel over every item that required a judge (Section 4.11), which returns $\alpha = 0.527$. We report this as a departure from the plan rather than as a threshold met.

All prompts were authored in English and, for a stratified subset of countries, additionally in a relevant native language (Supplementary Section S3). Native-language renderings were produced by an LLM translator (Claude Sonnet 4.6) prompted to preserve meaning, persona frame, proper nouns and acronyms, with the factual ground truth held in English since the target value is language-invariant. We did not employ human native-speaker translators, which remains a limitation (Section 5.7); we did audit the renderings after the fact by back-translating all 90 native prompts with a different model and having two further models compare each against the English original (Section 4.4).

3.9. Ground truth and the ground-truth registry

Ground truth is a set of four task-specific registries, released with the data, and we describe each as it was built rather than as it was planned (Supplementary Table S14 records the deviations). For T1 the key is the annual $\text{PM}_{2.5}$ value in force during the collection window, taken from the national instrument itself (statute, decree, regulation or gazette) and stored with the source, a verbatim excerpt and a status flag. Twenty-one countries have a numeric key. Three (Angola, Argentina, Nigeria) have no national $\text{PM}_{2.5}$ standard, so the correct answer is to say so, and a stated value is scored as fabricated. Bangladesh is scored against the 2005 gazette value because its 2022 revision could not be confirmed in an accessible official text, and Egypt has no key at all, because the official Arabic text could not be retrieved; Egypt is therefore excluded from T1 (56 cells). Where an instrument sets a ladder of staged values (Brazil’s five CONAMA stages, the EU limit of 25 with 10 from 2030, the United Kingdom’s 20 with an English target of 10 by 2040, the United States’ 9.0 succeeding 12.0), the registry stores the whole ladder as well as the value in force, so that citing a neighbouring stage can be separated from fabrication (Section 4.9). Canada’s CAAQS and the Philippine value are objectives rather than legally binding limits; they are the only national reference values those countries publish and are treated as the key, which the text notes where it matters.

For T2 the key is the most recent annual mean $\text{PM}_{2.5}$ concentration that the WHO Ambient Air Quality Database (v6.1, 2024) records for the prompted city, which compiles the values reported by national and subnational agencies; any reported year is accepted within $\pm 20\%$, so a model cit-

ing an agency value from a different year is not penalised for the year. Two cities have no $\text{PM}_{2.5}$ entry (Luanda, Cairo) and those cells fall to the judge. For T3 the key is the WHO Global Health Observatory estimate of deaths attributable to ambient air pollution (indicator `AIR_41`, 2021) with its uncertainty interval widened by half on each side as the accepted range, plus the published ordering of causes; the indicator covers ambient air pollution as a whole while the prompt asks about $\text{PM}_{2.5}$, and a model citing a different burden study (the GBD, a national cohort) is scored by the judge on whether its figure and sources are defensible, not against the WHO number. For T4 the reference set is the UNEP Global Air Pollution Legislation catalogue for each country, with manual overrides where it is stale. The registry records source, URL, access date and status per entry, but it has no human-validator field and no per-document hash: verification was documentary, by the authors, against the primary text. Entries without an equivalent official source are excluded from the code-adjudicated comparison for that cell, so missing ground truth is never scored as a model error.

3.10. Data collection

Calls were executed at temperature 0.3 with a fixed seed where the provider honours one (the three Ollama-served models), with 2 replications per prompt-model-persona combination, inside a fixed window to minimise model-update confounds; the reported accuracies are therefore point-in-time estimates. Two exceptions are declared: the GPT-5 family rejects any temperature other than the default, so GPT-5 and GPT-5-mini ran at temperature 1.0; and no provider offers a setting that makes output deterministic, so two replicates at 0.3 bound, rather than remove, within-model variance

(Supplementary Material compares the headline effects with that variance). Every response was stored with complete request metadata — pinned tag, timestamp, token counts, finish reason, persona condition — under SHA-256 checksums. Quality gates excluded truncated responses and responses in a language divergent from the request, both retained and analysed separately.

Per-cell coverage is not uniform: free and rate-limited endpoints returned fewer admissible responses than metered APIs, so the number of scored English-prompt cells ranges from 261 to 500 per model (777 for the model with the most native-language pairs when those are counted). We report each model’s realised N alongside its score (Table 2) rather than silently dropping under-covered cells, and the full coverage matrix, with quota and exclusion reasons for every shortfall, is in Supplementary Material.

3.11. Measures

Scoring assigns each task the most reliable instrument it admits.

Where the answer is a number in an official register, code decides. For T1, T2 and T3 the returned value is extracted and compared with the registry by a deterministic scorer, with plausibility guards that reject implausible extractions — a four-digit year read as a death count, a national population read as a mortality figure, a 24-hour value read as an annual one. The extractor normalises Devanagari digits, decimal commas and units written out in words, and its behaviour on the cases that matter for the language comparison (a comma decimal, a Hindi numeral, a parenthetical daily value beside the annual one) is fixed by a unit-test suite released with the code; a stratified manual audit of native-language responses it classified as carrying no value is reported in Supplementary Material. Comparing a value with

a key is exact, and a language model asked to do the same arithmetic can only reproduce it with noise. Code resolves 99.8% of the T1 cells in the analysis (1,614 of 1,617), 62.5% of T2 and 83.6% of T3; a T1 response with no extractable value counts as not having returned it. For the three non-standard countries, code scores whether the answer states that no national standard exists (correct), asserts a value (fabricated) or declines to answer (neither), and the primary tier comparison is reported with and without them.

Where judgement is required, a three-vendor panel decides by its mean. The residual of T2 and T3, with all of T4, is scored by Gemini 2.5 Pro, Claude Sonnet 4.6 and DeepSeek-V3, chosen for vendor diversity so that intra-vendor correlation does not inflate apparent agreement. T5, the open recommendation, retains the scores of the original collection judge; that judge was GPT-5-mini for the fifteen pre-specified countries and Claude Haiku 4.5 for the ten added in the extension, and because the extension holds seven of the ten Global North countries, judge and tier are confounded in any effect that still rests on those scores. This is why T1 was moved to code: on the original scores the Global South deficit in T1 was 0.21 against 0.38 under GPT-5-mini but 0.37 against 0.44 under Claude Haiku, a difference between judges as large as the difference between tiers (Supplementary Material). T5 carries no tier effect in either instrument. Reliability is reported as Krippendorff’s α , the single-judge ICC(2,1) and, as the operative quantity, the ICC(2,3) of the panel mean — a two-way random-effects, absolute-agreement coefficient, so that the judges’ different severities count against it — computed on every item the panel scored rather than on a calibration subset. One

panel member (DeepSeek-V3) is also an evaluated model, and we test for self-favouring directly. No human gold-standard layer was collected; ground truth is anchored instead to official primary sources (Section 3.9).

The unit of analysis is the scored cell, not the log line. Duplication exists at two levels and each has a stated rule. At the response level, 21.9% of (model, prompt, replicate) keys have more than one stored answer, because collection runs were resumed and re-run; the answer that enters the analysis is the first stored in run-file order, which is the one the judge scored, and the code verdicts are computed on that same answer so that the two instruments describe one response. Scoring every stored answer instead and averaging the verdicts changes the code verdict in 3.1% of cells and the mean by 0.0001 (Supplementary Material). At the score level, 951 cells (11.5%) carry more than one judge score, because the judge was re-run on parts of the set on different dates; those are repeated measurements of the same item, not independent observations, and are collapsed by their mean, leaving 8,300 cells, or 8,244 once the 56 Egyptian T1 cells without a key are removed (6,629 English-prompt). Treating re-runs as independent would be pseudo-replication: it inflates n and gives extra weight to whichever cells happened to be reprocessed. The choice is made on methodological grounds and does not depend on the result.

The primary composite combines five pre-specified subcomponents (Supplementary Section S6), each normalised to $[0, 1]$: factual accuracy against the registry entry (weight 0.30), contextual completeness (0.25), citation quality (0.15), calibration (0.15) and hallucination absence (0.15). Every headline effect is recomputed under equal weights and a PCA-derived weight-

ing, and a conclusion counts as robust only if its direction is stable across all three. The composite is computed per (country, model, persona) cell to support the H6 difference-in-differences test.

3.12. Analytical strategy

The three primary tests are pre-specified as a single family (F1) and corrected together with Bonferroni-Holm. Secondary tests (F2) are reported unadjusted within family, and declared exploratory tests (F3) use Benjamini-Hochberg FDR at $q = 0.10$ (Benjamini and Hochberg, 1995), whose guarantee assumes independent test statistics.

H1 (geographic gradient, primary) regresses country-level mean composite accuracy on the Joshi/HDI gradient via Spearman ρ , pre-specified one-sided at $\rho \geq 0.55$, with a Mann-Kendall trend test as a mandatory non-monotonicity complement. *H4 (corpus mechanism, primary)* relates corpus representation to accuracy by partial Spearman ρ adjusting for HDI and log GDP per capita, with Wikipedia counts as the pre-specified proxy and Wikidata coverage as exploratory complements. *H6 (persona, primary)* is a country-level difference-in-differences, $\Delta_{\text{GS}} - \Delta_{\text{GN}}$, supported if the persona condition reduces the Global South gap by at least 0.05 on the composite, with inference by a 5,000-permutation country-level test. *H3 (secondary)* is a one-sided Mann-Whitney contrast (Cliff’s δ). *H2 and H5 (exploratory)* are reported descriptively under F3 correction.

Three cross-cutting analyses corroborate the tier gap: a Gaussian linear mixed model (`statsmodels` MixedLM, REML) with crossed random intercepts for country and model, adjusting for centred HDI; a Bayesian hierarchical re-estimation (`pymc` (Abril-Pla et al., 2023), weakly informative priors);

and E-values for robustness to unmeasured confounding, via the approximation $RR \approx \exp(0.91 d)$ that VanderWeele and Ding (2017) prescribe for standardised effects and whose assumptions it inherits. The corpus mechanism is probed with an exploratory mediation model (`semopy` (Igolkina and Meshcheryakov, 2020)), and the persona manipulation with a role-frame-acknowledgement check. Pre-specified robustness analyses appear in Section 4.12.

3.13. Study protocol and ethics

The analytic dataset, after quality gates but before model fitting, is frozen and hashed, so the confirmatory analysis can be audited against the locked specification rather than taken on trust. Release terms are stated at the end of the paper.

No personal data from end-users of LLMs were collected; the study analyses model outputs to publicly relevant policy questions. Provider terms of service were reviewed and respected, and responses are redistributed solely for research under fair-use and reproducibility provisions, excluded from any commercial application.

4. Results

Sample note. Results below are reported on the *25-country* sample. The pre-specified benchmark covered 15 countries and was extended post hoc by ten (six Global North references — the United Kingdom, Canada, Australia, South Korea, France, Italy — and four native-pair countries — Colombia, Chile, Portugal, Angola). Fourteen LLMs are scored across five tasks, two personas, and four languages (English plus a country-native language — Spanish, Portuguese, or Hindi — for the nine applicable countries, giving the within-country English/native pairs that test H2). The primary family recomputed on the pre-specified 15 countries alone is tabulated in Supplementary Material, and no pre-specified conclusion is revised on the extension alone; where the two samples differ on a hypothesis the paper turns on, we say so in the text. Scoring is adjudicated by code wherever the answer is a number checkable against an official register, and by the mean of a three-vendor judge panel elsewhere (Section 4.11); the T1 ground truth is anchored to the national instrument in force for 24 countries and has no key for Egypt, which is excluded from T1 (Section 4.9).

4.1. Sample and scoring

The analysed corpus comprises 8,244 scored cells on the $[0, 1]$ composite, of which 6,573 are English-prompt responses across the 25 countries; the remainder are the within-country native-language pairs used for H2. Scoring uses the most reliable instrument each task admits: code adjudicates against the official register wherever the answer is a published number, resolving 99.8% of T1, 56.3% of T2 and 71.8% of T3 cells, and the residual, with all of T4, is scored by the mean of a three-vendor panel. T5 retains the scores

of the collection judge. In total 6,434 cells carry a code or panel score.

The panel mean replaces the single-judge design of the original protocol, in which the collection judge (GPT-5-mini for the pre-specified fifteen countries, Claude Haiku 4.5 for the ten extension countries) was itself among the evaluated models and, because the extension holds seven of the ten Global North countries, was confounded with tier. Judges differ sharply in severity (mean composite 0.617 for Gemini, 0.451 for Claude, 0.379 for DeepSeek), so a single judge fixes an arbitrary level, and single-judge reliability is only $\text{ICC}(2, 1) = 0.56$ against $\text{ICC}(2, 3) = 0.79$ for the panel mean. One panel member (DeepSeek-V3) is also an evaluated model, so we checked for self-favouring: it scores its own outputs *more* harshly relative to the other two judges (-0.177) than it scores any other model’s (-0.123 to -0.160).

4.2. Accuracy by model

Table 2 orders the 14 models by mean composite: frontier systems lead and small open-weight models trail. On H3, the Brazilian-Portuguese regional model Cabra-Mistral 7B is the *weakest* of all (0.351 versus 0.557 for the rest; Mann-Whitney $p \approx 0$, Cliff’s $\delta = -0.48$). Cliff’s δ has a direct reading — the probability that a randomly drawn answer from one group beats one from the other, rescaled to $[-1, +1]$ — so -0.48 means the regional model loses roughly three such comparisons for every one it wins. That comparison is against the whole roster, which includes frontier systems; the pre-specified test was narrower, Cabra against its scale-matched peer Llama 3.1 8B on Portuguese-language prompts, and it points the same way with a smaller margin: on Brazil the regional model scores 0.258 against 0.313 ($\delta = -0.26$, $n = 20$ each), and across the three Portuguese-speaking

countries 0.304 against 0.362 ($\delta = -0.17$, $n = 60$ each). Regional instruction tuning at the 7B level does not recover what scale and frontier training give, even in the language and country it was tuned for, though with 20 Brazilian cells per model the narrow test is descriptive.

4.3. Accuracy by task: the recall floor

Difficulty is sharply task-dependent (Table 3). The hardest task by a wide margin is T1 (technical-standard recall, 0.394) — the task that requires returning the national annual $\text{PM}_{2.5}$ standard in force. The open recommendation task (T5) scores highest (0.769) because its rubric rewards plausible, well-structured policy advice that does not hinge on a single fact. Pooling the two recall tasks (T1, T2; mean 0.435) against synthesis and recommendation (T3–T5, 0.612), the difference is large (Mann-Whitney $p \approx 0$, Cliff’s $\delta = -0.41$, roughly two and a half losses for every win). LLMs are weakest exactly where applied environmental management needs them most: recalling the precise regulatory threshold in force.

The two corrections of the scoring are themselves informative here. T2 was originally scored against a placeholder key and rises from 0.368 to 0.473 once each returned concentration is compared with the authorised range by code: the correction moved the task whose key was defective. T1 always had an official key but was scored by two collection judges of different severity, and moving it to code raises it from 0.367 to 0.394 while keeping it the hardest task; the code verdict also shows what the judges could not, that a third of the wrong T1 answers (259 of 754) cite a value that sits on the country’s official ladder — a superseded value or a future stage — rather than a fabricated one (Section 4.9). The floor is real, shallower than first

reported, and concentrated in normative recall (T1).

4.4. H2 — asking in the local language lowers accuracy

For the nine countries with a target native language, each prompt is rendered in both English and the native language with content held fixed, so the contrast is within-country and within-model. Replicates are averaged within each (model, prompt) cell before pairing, since pairing replicate-to-replicate would inflate n and make the test anticonservative. Across the 839 matched cells, native-language prompting *lowers* mean composite accuracy by 5.0 percentage points (Wilcoxon signed-rank $p = 5 \times 10^{-16}$). The penalty tracks the resource level of the language, and all three are significant (Table 4): Hindi -11.4 pp, Spanish -4.5 pp, Portuguese -4.2 pp. India, which scores highest of all 25 countries in English, drops the furthest in Hindi.

Because H2 carries the study’s central claim, we tried to make it disappear rather than to confirm it, and it held under every attack: it is not driven by any one country (leave-one-out -4.4 to -5.5 pp, all significant), by any one model (-4.5 to -5.2 pp), or by outlying cells (5% trimmed -4.5 pp, $p = 6 \times 10^{-20}$); it survives every weighting of the composite; and it appears in every task and under both persona conditions.

Two attacks answer the objections a sceptical reader raises first. The first is that language-model judges might penalise non-English prose for style rather than content, so we test a binary outcome no judge touches: does the response contain a value the extractor can compare with the register? The extractor reads Devanagari numerals, decimal commas and units written out in words, so a Hindi or a Spanish-language number is not penalised for its notation (Section 3). English prompts yield a verifiable value 71.9% of the

time and native-language prompts 62.5%, a paired difference of -9.4 pp, with the native version worse in 134 of the 218 discordant pairs (sign test $p = 9 \times 10^{-4}$): Hindi -39.5 pp, Spanish -13.4 pp, while Portuguese, the best-resourced of the three, shows no such loss ($+3.5$ pp). Asking in the local language does not merely lower a judged score; where the language is scarce it makes the model substantially less likely to return a checkable number at all.

The second is that the native prompts, produced by an LLM translator, might simply ask something different. We tested this on all 90 unique native prompts — a census, not a sample — back-translating each with a *different* model and having two further models audit the result against the English original. On the two items that could confound the effect, fidelity is complete: no prompt changed the country and none changed the requested quantity (90/90 on both). Restricting H2 to the prompts verified as faithful leaves it essentially unchanged (-4.8 pp, $p = 2 \times 10^{-8}$) against -5.2 pp among the divergent ones, a difference not distinguishable from zero (permutation $p = 0.80$). Translation fidelity does not predict the size of the penalty, which is what one expects if translation is not causing it. Separating the pairs by the instrument that scored the native cell, the penalty is -3.8 pp where code decided ($n = 333$, $p = 0.004$), -8.6 pp where the panel decided ($n = 314$, $p < 10^{-21}$) and -1.1 pp, not significant, where the collection judge’s score remains ($n = 192$, essentially T5, the open recommendation with no register value). The code-adjudicated subset is the smaller of the two because code acts only where a number exists, so it conditions on the model having produced a value in *both* languages and selects away the severest

failure mode; the panel subset carries that failure mode, which is why judge leniency cannot be assumed to cancel across languages and why the judge-free outcome above is the one that settles the point.

The applied implication is the opposite of the intuition. A manager who suspects an English-language model serves their country poorly, and responds by asking in the national language, makes the answer *worse* — and worse by the largest margin precisely where the language is least represented in training corpora.

4.5. *H6 — persona does not narrow the geographic gap*

The persona factor was fully crossed, so H6 is tested as a difference-in-differences. The public-environmental-manager frame leaves the North/South gap essentially unchanged: +4.8 pp under the neutral frame versus +5.2 pp under the manager frame, a DiD of +0.5 pp, far below the pre-specified 5 pp threshold; a country-level permutation test (5,000 permutations of the tier label) gives $p = 0.29$. H6 is not supported: presenting the query as coming from a public environmental manager neither narrows nor amplifies the Global South disadvantage, consistent with evidence that expert personas do not reliably improve factual accuracy (Zheng et al., 2024; Basil et al., 2025). A manipulation check finds that 50.2% of persona-condition responses explicitly acknowledge the role frame, so the null is not an artefact of the frame being universally ignored.

4.6. *H1 — a tier gap that holds, a gradient that does not*

H1 was pre-specified in two parts, and they separate cleanly.

The ten Global North countries average 0.572 against 0.521 for the fifteen Global South countries, a gap of +5.0 percentage points whose 95% bootstrap confidence interval over countries is $[+1.5, +8.4]$ pp and *excludes zero* (permutation of the tier label, country as unit, $p = 0.021$). It is stable to dropping any single country (leave-one-out $[+4.4, +5.9]$ pp) and to the composite weighting (+4.2 pp under equal weights, +8.7 pp on the factual subcomponent, where it widens). Because seven of the ten Global North countries entered with the extension, we also report the gap where the wave cannot explain it: within the extension alone (seven Global North against Colombia, Chile and Angola) it is +5.9 pp $[+2.3, +9.6]$, and on the pre-specified fifteen (three Global North against twelve) +6.0 pp $[+1.7, +10.2]$, a point estimate of the same size on a sample too small to make it significant by permutation ($p = 0.13$). The gap is a property of the data, not of the wave that collected it.

Where the gap lives, and why the composite understates it. The composite averages five tasks and the gap is not spread evenly across them. Re-estimating the same contrast as a binomial mixed model on each task separately — outcome coded as returning the register value, with crossed random intercepts for country and model — shows the disadvantage scaling with how much the task depends on a fact specific to that country (Table 5). On the binding national standard (T1), scored by code against the instrument in force, the Global South odds of returning the register value are 0.33 of the Global North’s (50.7% versus 31.6% raw), and on policy instruments (T4) also 0.33; on health-evidence synthesis (T3), where the answer is an international estimate rather than a national one, the ratio rises to 0.69;

and on open recommendation (T5), the one task with no register value to get right, the gap disappears entirely. The T1 ratio survives the two objections its composition invites (Section 4.9): setting aside the four Global South countries whose key is not a plainly verified national value it is 0.39 [0.31, 0.50], and on the pre-specified fifteen it is 0.27 [0.21, 0.34]. The gap is not weak — it is *concentrated*, and the composite dilutes it by averaging tasks where it is severe with a task where it does not exist. On the question a public manager most often brings to these systems, what is the standard in force here, a Global South country is served at a third of the odds of a Global North one.

At $n = 25$, Spearman $\rho(\text{accuracy}, \text{HDI}) = +0.36$ does not reach conventional significance ($p = 0.076$; Mann-Kendall $p = 0.065$) and stays well below the pre-specified $\rho \geq 0.55$ threshold, which is the criterion H1 was fixed against. At the pre-specified $n = 15$ it is essentially absent ($\rho = +0.08$), and no leave-one-country-out re-estimate reaches 0.55 either (range [+0.30, +0.48]); under equal weights it is 0.35 and under PCA weights 0.37. We therefore report a tier difference we can demonstrate and a development gradient we cannot: the disadvantage is established as a difference between groups, while its shape across the development range remains exploratory. Under Bonferroni-Holm across the primary family {H1-gradient, H4, H6}, no test rejects its null.

The country ordering retains large axis-orthogonal heterogeneity (Supplementary Material): India (0.651, Global South) tops all countries, Indonesia and South Africa outrank most of the Global North, and Canada and France sit low. The disadvantage is a tier difference layered over substantial country-

specific variation, and that heterogeneity is itself part of why the gradient does not resolve.

4.7. H4 — corpus representation: country coverage, inseparable from development

The pre-specified mechanism was training-corpus representation, proxied by Wikipedia article and edition count. That proxy is null: Spearman $\rho = +0.05$ ($p = 0.82$) uncontrolled, and still null partialling out HDI and log GDP per capita. A post-hoc country-coverage proxy reaches $\rho = +0.37$ ($p = 0.070$) and attenuates after adjusting for HDI (partial $\rho = +0.19$, $p = 0.37$). At country level neither channel is established, because with 25 countries coverage and development are collinear.

Changing the unit of variation resolves what the country level cannot

The obstacle is confounding rather than absence of signal, and the way out does not require more countries. Each country answers five tasks that differ in how much the answer depends on a fact about *that* country: T1, T2 and T4 do; T3 (an international WHO estimate) and T5 (a generic recommendation) do not. Countries score 0.462 on the first block and 0.658 on the second, a specificity deficit of 0.195. If what limits the model is how much the corpus says about a given country, that deficit should be smaller where coverage is greater. Estimating the interaction between coverage and task dependence at the response level, with a fixed effect for country, gives $\beta = +0.030$ (SE 0.005, $p = 9 \times 10^{-10}$, $n = 6,573$). The fixed effect absorbs every country-level *main* effect — development, income, official language — but not their interaction with task dependence, and that is where the rival lives: a richer country may

simply have more of its regulation online, feeding both the model’s training data and the ease of building the key.

Four checks separate the coverage signal from artefact (Table 6). The *purest* contrast — T1, wholly national, against T5, wholly generic — yields the *largest* coefficient; dropping T5 or T4 leaves it intact; a placebo inverting the labels returns the exact mirror image; and entering the language-corpus proxy alongside it, coverage survives ($\beta = +0.030$, $p = 2 \times 10^{-9}$) while language does not ($\beta = +0.004$, $p = 0.45$). Because every response here was elicited in English, that last result was predicted and is the falsifiable part of the design. The rival it does not eliminate is development. Coverage and HDI correlate at $\rho = 0.65$ across the 25 countries, and when the two interactions enter the same model the ordering reverses: HDI $\times$ dependence survives ($\beta = +0.027$, SE 0.007, $p = 5 \times 10^{-5}$) and coverage $\times$ dependence does not ($\beta = +0.012$, SE 0.007, $p = 0.063$). Within the nine anglophone countries, where the language of the question is the same by construction, coverage predicts the deficit at the full magnitude ($\beta = +0.024$, $p = 0.003$); that removes language as the explanation, not development.

The resulting claim is therefore narrower than “coverage causes the gap”, and narrower than we first wrote it. What the within-country design establishes is that the country-specific deficit is smaller where the world records more about the country — whether that record is measured as encyclopedic coverage or as development, which in this sample are two readings of the same variable. It rules out the language-corpus channel; it does not pick coverage over development, and we report the mechanism as identified only up to that pair.

The corpus account shows itself more cleanly in H2, where the unit is the language: the per-language penalty falls monotonically with mC4 token volume (Xue et al., 2021) and OSCAR size (Abadji et al., 2022), with Hindi, the smallest corpus, carrying the largest penalty — though with three languages this remains descriptive.

4.8. *H5 — open versus closed-accessible*

Closed-accessible models average +13.0 pp above open-weight ones, against the optimistic H5 framing. The open arm includes larger models, so the gap is not attributable to scale alone, although DeepSeek-V3 ranks first overall. H5 is interpreted descriptively.

4.9. *Ground-truth provenance, and what the T1 key decides*

Every key comes from an official register, anchored by documentary verification and reproducible from the cited URLs: T1 from the national regulation, T2 from WHO Ambient Air Quality Database v6.1, T3 from WHO GHG indicator AIR_41, T4 from the UNEP Global Assessment of Air Pollution Legislation. Three countries have *no* national PM_{2.5} standard, confirmed from the official text — Nigeria regulates PM₁₀ only, Argentina has no binding federal value, Angola no national legislation — which is where a faithful model must decline to state a non-existent threshold. Two keys are not fully verified (Bangladesh, scored against its 2005 gazette; Egypt, with no retrievable value and therefore no T1 key), and all five of these cases are Global South, so the composition of the key is itself a candidate explanation for the T1 gap. It is not: set them aside and the Global South odds on T1 are 0.39 [0.31, 0.50] of the Global North’s, against 0.33 with them. The

full registry is in Supplementary Material. Wikipedia and Wikidata appear only as the explanatory variable of the mechanism test; no answer is scored against them.

The key in force, and the ladder around it. Several instruments set a ladder of values rather than one: Brazil’s CONAMA 506/2024 runs 20, 17, 15, 10, 5 $\mu\text{g}/\text{m}^3$ across five stages, of which 17 has been in force since January 2025; the European limit is 25 with 10 from 2030; the United Kingdom’s limit value is 20 with an English target of 10 by 2040; the United States moved from 12.0 to 9.0 in 2024. The T1 key is the value in force in the collection window, and it is exacting: no model returns 17 for Brazil (the modal answer is 15, the next stage), almost none returns 20 for the United Kingdom (most return the 2040 target or the superseded 25), and fewer than a quarter return 9.0 for the United States. Of the 754 wrong T1 answers, 259 cite a value that sits somewhere on the country’s official ladder — a stage not yet in force or a value since replaced — which is confusion about *which* value binds, not fabrication, and is the failure a training cut-off would produce. The distinction does not rescue the tier comparison, it sharpens it: scoring any ladder value as correct raises Global North accuracy on T1 from 0.51 to 0.77 and Global South accuracy only from 0.32 to 0.40, because the staged instruments are mostly in the North, and the odds ratio falls to 0.12 [0.10, 0.15]. Where the register offers several plausible numbers the models know them; where it offers one, they often do not.

4.10. Qualitative failure modes

Three failure modes recur behind the composite, with verbatim examples in Supplementary Material. Models *fabricate standards that do not exist* for

the three countries with no national norm: in 168 of the 256 scored T1 cells for Angola, Argentina and Nigeria the answer asserts a specific national value, in 33 it correctly states that none exists, and in 31 it declines to answer, and the invented values disagree among replicates; they *miss the binding value*, answering $15 \mu\text{g}/\text{m}^3$ for Brazil where the stage in force is 17 (CONAMA 506/2024), a confusion of stage rather than an invention; and they *degrade in the native language*, as when GPT-OSS 120B returned India’s correct $40 \mu\text{g}/\text{m}^3$ in English and an *empty* answer in Hindi.

4.11. Inter-judge reliability

Reliability is reported on the sample that produces the effects, not on a calibration subset. All 3,190 items requiring a judge were scored by all three panel members, giving Krippendorff’s $\alpha = 0.527$, single-judge $\text{ICC}(2, 1) = 0.558$, and — the operative quantity, since the analysis uses the panel mean — $\text{ICC}(2, 3) = 0.791$.

The panel also isolates a design rule that generalises beyond this study. Where the judging prompt asked the model to compare a returned concentration against a tolerance band expressed in words — that is, to perform the arithmetic itself — inter-judge concordance on those items was $r = -0.04$. Pre-computing the admissible range and asking only whether the value falls inside it raised concordance on the same items to $r = +1.00$, and the 274 scores collected under the ambiguous wording were discarded and recollected. Apparent judge disagreement can be an artefact of what the judge is asked to compute. Arithmetic belongs to code.

4.12. Robustness

The two supported effects were tested against seven attacks and survive all of them; the development gradient stays below the pre-specified 0.55 under every one of them (its highest value, 0.48, is a leave-one-out re-estimate), which is what makes its exploratory status a finding rather than an omission. Full outputs are in Supplementary Material.

No single country, and no weighting, carries the result. Leave-one-country-out re-estimation holds the tier gap in $[+4.4, +5.9]$ pp and the native-language penalty at -4.4 to -5.5 pp. Under equal weights the gap is $+4.2$ pp and the language penalty -4.8 pp; restricting to factual accuracy against the register value widens them to $+8.7$ and -6.1 pp and deepens the recall floor, so both effects grow where the measurement is most objective. The two replicates of each cell differ by 8.5 pp on average (within-cell SD 10.1 pp), so the -5.0 pp language penalty, averaged over 839 pairs, stands roughly ten standard errors from the noise two replicates leave unresolved.

The tier gap survives a distribution-free test and two model families. A permutation test shuffles the North/South labels among the 25 countries ten thousand times and counts how often chance alone produces a gap this large; it assumes nothing about the distribution and treats the country, not the answer, as the independent unit. Chance clears the observed gap 2.1% of the time ($p = 0.021$). A Bayesian hierarchical model with crossed random intercepts agrees ($\beta = -0.048$, 94% HDI $[-0.090, -0.009]$, $P(\beta < 0) = 0.987$), while a frequentist mixed model returns the same magnitude and sign but leaves it marginal ($\beta = -0.063$, $p = 0.087$), because it charges the large between-country heterogeneity to the error term. Two of three

tests clear zero and all three agree on magnitude, so we report the gap as supported and specification-sensitive, with an E-value of 1.70 at the point estimate and 1.30 at the confidence limit nearest the null ([VanderWeele and Ding, 2017](#)) — a country-level confounder of modest strength would suffice to explain it away.

The gap tracks world-system position, not national wealth. Re-estimated under three development-based partitions of the same 25 countries, the gap is significant under UNCTAD and non-significant under every split drawn on HDI (Supplementary Material). This coheres with the rest of the paper: the HDI gradient is itself null, so a split on HDI should not be expected to produce a gap, and that UNCTAD separates the data where HDI does not points to position in the world system rather than income as the operative variable ([Quijano, 2000](#); [Mohamed et al., 2020](#)). The reading is post hoc, so we claim the gap for the North/South distinction as conventionally drawn, not as a general statement about development.

What the null gradient does and does not exclude. Simulating 25-country samples at known correlations, the design has 77% power at $\rho = 0.55$ — the threshold fixed in advance — and a minimum detectable effect of $\rho = 0.56$. A gradient of the magnitude we committed to calling substantial is excluded; a weak one is not (power 48% at $\rho = 0.40$). The ten countries added post hoc fall *within* the HDI range the original fifteen already spanned, so the extension raises the density of observations rather than the range of the predictor.

4.13. Summary

Table 7 maps each hypothesis to its finding, the candidate mechanism, and an explicit evidence status, so no claim is read as stronger than the data support. Three effects are supported and survive every robustness attack in Section 4.12; two pre-specified expectations are not met; and the corpus mechanism is identified within countries only up to the pair coverage-or-development, which this design cannot separate.

5. Discussion

We set out to measure whether LLMs answer air pollution policy questions as reliably for the Global South as for the Global North, whether prompt framing changes the answer, and what mechanism drives any gap. Across 25 countries, 14 models, four languages, five tasks and two personas, scored against official ground truth by code where the answer is a register value and by a three-vendor panel elsewhere, the harms we can demonstrate are practical rather than aetiological: the three remedies an agency would reach for all fail, the most intuitive of them makes the answer materially worse, and models are least reliable on the recall of the binding value in force. A North/South tier difference persists, and we identify the channel behind it only where the design has the resolution to do so.

5.1. Three intuitive remedies, all failing — and one that harms

The local language does not help and materially hurts. Asking in the country’s own language lowers accuracy by 5.0 pp, significantly in all three languages, and the penalty scales with the scarcity of the language’s corpus,

with India — the highest-scoring country of all 25 in English — falling furthest in Hindi. This is the mechanism the resource hierarchy predicts (Joshi et al., 2020; Xue et al., 2021), and the measurement has the resolution to see it because the unit is the language rather than the country. The effect depends on no single country, model, task, persona condition or scoring instrument, nor on translation quality, and it holds on an outcome involving no judge at all: the model becomes markedly less likely to produce a checkable number in the first place.

The local-manager persona does not narrow the gap (DiD +0.5 pp, permutation $p = 0.29$), consistent with evidence that expert personas do not reliably improve factual accuracy (Zheng et al., 2024; Basil et al., 2025). This null carries more weight than a null about a folk habit would, because role framing is the first thing every major vendor tells its users to do — Anthropic, OpenAI, Google and xAI converge on the same advice (Section 2), so an administration following its supplier’s documentation writes the sentence we tested. What that documentation claims for the sentence is control over tone, format and framing, and nothing in it marks the boundary: nothing tells a reader that the technique which reliably shapes register does not also shape factual reliability. Our estimate separates the two. A manipulation check confirms half the responses explicitly take up the role, so the frame is received and changes the voice; it does not change what the model knows. We tested it typed into the message rather than installed as a system instruction, the channel a public servant in a chat interface actually has (Section 5.7).

The regional model is the *worst* performer of all fourteen ($\delta = -0.48$), and it also trails its scale-matched peer on the Portuguese-language Brazilian

prompts it was tuned for (0.258 against 0.313 for Llama 3.1 8B, on 20 cells each), so reaching for a locally tuned small model trades away the scale and frontier training that carry domain accuracy without buying back the local knowledge it promises; the narrow test is small, and we weight the claim accordingly.

Practitioners deploy all three as fixes; none works, and the one a manager is most likely to try first does active damage. For Global South environmental agencies, no cheap prompt- or model-level workaround substitutes for verifying the binding value against the official source.

5.2. The recall floor sits on the values that bind

Accuracy collapses on the tasks that require returning a specific value in force: 0.435 for normative and local-datum recall against 0.612 for synthesis and recommendation ($\delta = -0.41$), deepest on the national standard itself (0.394). The risk falls unevenly: a manager drafting prose about air quality policy is comparatively well served, a manager retrieving the regulatory cutoff a decision turns on is in the worst cell of the design, and worse still if they ask in a lower-resource language. The floor is genuine rather than an artefact of scoring: both recall tasks are now decided by code against the register, T2 rose from 0.368 to 0.473 when its placeholder key was rebuilt, and T1, once moved from two collection judges of different severity to code, stays the hardest task at 0.394, with a third of its wrong answers citing a value from the country’s own official ladder rather than an invented one.

5.3. *A tier difference we can show, and a gradient we cannot*

H1 was pre-specified in two parts, and the corrected scoring separates them: the tier gap holds and the monotonic gradient does not.

Read on the composite the gap looks modest (+5.0 pp, bootstrap CI [+1.5, +8.4]), and it is the same size inside the extension wave alone and on the pre-specified fifteen, so the wave that collected it does not explain it. Estimated task by task on the binary outcome of returning the register value, it is not modest: the gap scales with how much the answer depends on a fact specific to that country, from an odds ratio of 0.33 on the binding national standard — 0.39 once the countries with no or unconfirmed standard are set aside — to no gap at all on open recommendation, the one task with no register value to get right. The disadvantage is concentrated, and averaging five tasks into one composite dilutes it with tasks where it does not exist. On the question a manager actually brings to these systems — what is the standard in force here — a Global South country is served at a third of the odds of a Global North one. That is the faithful statement of the geographic finding.

The gradient stays below the pre-specified threshold on every reading, and the null is informative rather than empty because the design has 77% power at the pre-specified $\rho = 0.55$: a gradient of the magnitude we committed to is excluded, while a weak one is not. The country ordering shows why the two results differ. India tops all 25 countries despite being Global South; South Africa and Indonesia outrank most of the Global North; Canada and France sit low. The disadvantage is a tier difference layered over substantial country-specific idiosyncrasy, and that idiosyncrasy is what a rank correlation over

25 countries lacks the resolution to see through.

5.4. *The mechanism, where the design has resolution*

The pre-specified mechanism test fails, and the reason is a defect in the instrument rather than an absence of signal. The proxy assigns the size of the Wikipedia edition in a country's dominant official language, so nine of our 25 countries receive the value of the *English* Wikipedia. The variable does not measure how much text exists in a country's language; it measures whether the country has English as an official language, which is a fact about colonial history. We report this because it bears on any study that has used or will use the same proxy.

Partial resolution comes from changing the unit of variation. Within the nine anglophone countries the linguistic channel is not merely controlled but identical by construction, and country coverage still predicts the specificity deficit ($\beta = +0.024$, $p = 0.003$): what that coefficient measures cannot be language. What it can be is development. Entered together with coverage, HDI $\times$ task dependence takes the signal ($\beta = +0.027$, $p = 5 \times 10^{-5}$) and coverage falls to $\beta = +0.012$ ($p = 0.063$), because across these 25 countries the two correlate at $\rho = 0.65$ and the design has no lever that moves one without the other. The mechanism is therefore identified only up to that pair: the deficit is smaller where more about the country is on record, and whether the operative record is the encyclopedic footprint or the wider apparatus of a developed state this study cannot say.

5.5. *The methodological lesson: the key matters more than the sample*

The same scored cells support different substantive conclusions depending on the instrument used to score them, and the size of that difference is the lesson. Scoring two of the five tasks against placeholder ground truth, and delegating to a language model the arithmetic of deciding whether a returned number fell inside an authorised range, yields a development gradient of $\rho = 0.51$ ($p = 0.004$) and a native-language penalty of -2.1 pp. Rebuilding those keys from official registers and moving the comparison into code cuts the gradient to $\rho = 0.36$, below the pre-specified threshold, more than doubles the language penalty to -5.0 pp, and converts Spanish from a null into a significant harm. Moving the third register-valued task, T1, from a judge to code changed the headline odds ratio on that task from 0.22 to 0.33, because the two collection judges had been split along the same line as the tiers. The dataset did not change; only the measuring instrument did. The field’s instinct when a geographic-bias effect proves fragile is to add countries, but the ground truth deserves the scrutiny first: a defective key does not merely add noise, it adds *structured* noise, because the defects concentrate wherever official data are hardest to obtain — which is exactly where the hypothesis lives.

5.6. *Why the domain matters*

Fine particulate matter is among the leading environmental risk factors for premature mortality, and the burden falls disproportionately on the Global South: the World Health Organization attributes 88,548 deaths in Brazil in 2021 to ambient air pollution (uncertainty interval 69,477 to

108,431) ([World Health Organization, 2026](#)), and short-term exposure continues to drive measurable mortality in Brazilian cities ([Requia et al., 2024](#)). When a manager delegates a normative lookup or an episode recommendation to an LLM, an inaccurate answer can propagate into a regulatory brief or an emergency response.

We do not claim the model is least reliable where consequences are greatest, because we did not measure it: disease burden is not among our country covariates, and the one country-level observation we have points the other way, since India carries one of the largest PM_{2.5} mortality burdens in the world and is also the highest-scoring country in our sample. Whether reliability is misaligned with need is answerable by adding an age-standardised attributable-mortality covariate, but not by this design. We interpret the pattern we *did* measure through the coloniality-of-knowledge framework ([Mohamed et al., 2020](#)): LLMs are compressed summaries of corpora produced under uneven digital and publication access, and the language penalty is a measurable link from corpus asymmetry to knowledge asymmetry.

5.7. *Limitations and future opportunities*

Each limit below bounds this design, and each names the study that would remove it. We list them in the order we would run them.

Role framing was tested in the user turn, not the system prompt. The manager frame is prefixed to the message rather than installed as a system instruction, where three of the four vendor guides place it — Google’s admits either that slot or the start of the user prompt, which is the one we used. The choice follows the population we care about — public servants in the chat interface, who have no system prompt to configure — but our null

bounds that user’s channel, not a developer’s. Whether a system-instruction persona recovers factual accuracy where a typed one does not is a clean two-cell experiment on the same items, and it is the first study we would run.

From panel-scored to gold-anchored. No human expert re-scored responses here. Three properties bound what that costs: where the answer is a register value the verdict is computed by code; where judgement is required the score is the mean of three vendors rather than one; and every reported effect is a between-group contrast, in which a judge’s systematic severity cancels. Restricting to objective factual accuracy *strengthens* both headline effects, the opposite of what a permissive judge would produce. A blinded human-expert validation of a stratified sample — roughly 150 responses scored by three raters, which the released pipeline exports directly — would convert an official-source-anchored result into a gold-standard-anchored one.

From a within-country signal to a causal design. Within countries we remove country-level confounding without removing task-by-country confounding. A design that varies country coverage while holding development fixed, or a longitudinal study tracking accuracy as a country’s encyclopedic footprint grows, would close that gap.

From a null gradient to an adequately powered test. The tier difference is established while the gradient is not, at $n = 25$. A pre-registered replication at substantially larger n , which the open dataset makes cheap, would settle whether the gradient is genuinely absent or still underpowered.

From three native languages to the full multilingual matrix. The strongest result rests on three native languages, so its *mechanism* is reported descrip-

tively even though the effect is not. The translation-quality objection is settled empirically by the back-translation census reported above; what remains open is breadth, not validity, and scaling to more languages is the highest-value extension this study points to.

From one domain to a cross-domain program. All effects come from one high-stakes applied domain and one corpus family (Wikimedia), and we confine the mechanistic language accordingly. The same task suite and ground-truth-registry method transfer directly to other regulated, source-anchored domains — water quality, food safety, vaccine policy.

Served-model stack as the unit of analysis. We evaluate models *as served* through fixed access stacks, which is what a public manager queries; a residual gap could in principle reflect the stack rather than the weights. A controlled quasi-isolation design with identical open weights across stacks would separate the two, which the per-call provenance we release already enables.

5.8. *What this study contributes*

The three fixes practitioners reach for first fail together: the local language actively harms, the persona frame does nothing, and the regional model is the single worst performer of fourteen. A benchmark’s most useful output is often telling the field what does not work, with effect sizes behind it. The methodological result is that in this literature the ground-truth key, not the sample size, is the binding constraint. And the design separates what 25 countries can demonstrate from what they cannot, reporting the second set as open rather than as supported.

6. Conclusion

Large language models are becoming working infrastructure for air pollution policy, a domain where an inaccurate answer about a binding standard, a measured concentration, or an operational response carries direct public-health stakes in the Global South, where the $\text{PM}_{2.5}$ mortality burden is heaviest ([GBD 2019 Risk Factors Collaborators, 2020](#); [Requia et al., 2024](#)). We measured geographic and persona-modulated bias in that domain with a 25-country, 14-model, four-language benchmark (English plus the native language of nine countries), scored against ground truth anchored to official primary sources — by code wherever the answer is a register value, and by the mean of a three-vendor judge panel elsewhere.

The remedies an agency can try do not work, and the most intuitive one backfires. Asking in the country’s own language *lowers* accuracy by 5.0 percentage points, most sharply where the language is least represented in training corpora; a local-manager persona does not narrow the gap; and the regionally tuned model is the weakest of all fourteen systems. Accuracy also collapses on the recall of a binding value, the task a manager can least afford to have wrong. A Global North/South tier gap persists and is highly structured: on returning the binding national standard the Global South odds of a correct answer are 0.33 of the Global North’s, and the gap shrinks as the answer depends less on a country-specific fact, disappearing on the one task with no register value to get right. The monotonic development gradient does not reach significance, so we report a difference between groups that we can demonstrate and a shape across the development range that we cannot. On mechanism, the within-country design rules out the language-

corpus channel for a gap measured in English, but cannot separate country coverage from development, and we claim no more than that.

For Global South environmental agencies the practical message is cautionary. LLM outputs on binding standards and measured data should be treated as drafts checked against the official gazette, and none of the cheap prompt- or model-level workarounds substitutes for that verification — least of all switching to the local language, which on this evidence makes the answer worse. We read the pattern through the coloniality-of-knowledge framework (Mohamed et al., 2020): the penalty falls on the languages underserved in training corpora (Joshi et al., 2020), and it falls hardest on Hindi, the scarcest of the three; whether those languages are also spoken where the health burden of air pollution is greatest is not something this design measures, and the one country-level observation we have (India) points the other way (Section 5).

One methodological result generalises beyond the domain. Rebuilding two task keys from official registers and moving range comparison out of the judge and into code changed the substantive conclusions on the same responses. When an effect in this literature proves fragile, the ground-truth key deserves the audit before the sample does, because a defective key adds *structured* noise that concentrates exactly where official data are hardest to obtain.

Protocol, prompts, official-source ground-truth registries, analysis code, and anonymised response data are released openly, with a one-command reproduction pipeline and a method-audit gate that recomputes every reported number from the raw data. The study also provides a reusable template for

applied LLM-equity audits — official-source ground truth, code adjudication wherever the answer is a checkable number, and end-to-end reproducibility — in the high-stakes public-sector domains where these systems are now being deployed.

Data and code availability

The full protocol, prompts, analysis code, ground-truth registries, and the anonymised LLM responses will be released on publication at github.com/Roverlucas/artigo-vies-analise-regional under MIT (code) and CC-BY-4.0 (data, prompts, documentation) licenses. The release includes the per-response records that every reported effect is computed from, so each number in this paper can be recomputed from the raw data rather than taken on trust. The analysis pipeline runs end-to-end via `python code/run_all.py -confirmatory`.

Study protocol

Hypotheses, sampling frame, outcome definition, decision thresholds and multiplicity correction were fixed in a version-controlled analysis plan before confirmatory collection began. That plan was not deposited in a public registry, so readers cannot verify its timing independently of the repository history, and we do not claim pre-registration. Delivered scope also departed substantially from the planned design, which the plan itself made a condition for confirmatory status.

We therefore report the study in two layers. The first states what was specified in advance and what became of it: the three co-primary hypotheses

and the secondary hypothesis are reported against their declared thresholds, including the three that failed and the one whose registered direction the data reverse. The second reports what the data suggested and the plan did not anticipate, labelled exploratory throughout, with intervals and without confirmatory language. Deviations from the plan are tabulated in the supplement.

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Author contributions

Following CRediT taxonomy:

- Lucas Rover (UTFPR): Conceptualization, Methodology, Software, Data Curation, Formal Analysis, Investigation, Visualization, Writing – Original Draft, Project Administration.
- Vitor de Melo Dominski (Faculdade Descomplica): Data Curation, Software, Formal Analysis.
- Prof. Anibal Tavares de Azevedo (Unicamp): Writing – Original Draft, Writing – Review & Editing.
- Dr. Eduardo Tadeu Bacalhau (UFPR): Validation, Supervision, Writing – Review & Editing.

- Dra. Yara de Souza Tadano (UTFPR): Validation, Supervision, Writing – Review & Editing.

Declaration of generative AI in the writing process

Two distinct uses must be separated here, because only one of them is what this declaration is about.

As an object and instrument of the study. Large language models are the subject of this research, and they also serve as one of its measuring instruments: GPT-5-mini and Claude Haiku 4.5 scored model responses as collection judges, Gemini 2.5 Pro, Claude Sonnet 4.6 and DeepSeek-V3 re-scored them as a three-vendor panel, and Claude Sonnet 4.6 produced the native-language renderings of the prompts. These uses are part of the method, are described in Section 3, and are why the study also reports inter-judge reliability and a back-translation audit of the translated prompts. Wherever the correct answer is a value in an official register, the verdict is computed by code rather than by a model.

In preparing the manuscript. During the preparation of this work the authors used Anthropic’s Claude (via Claude Code) to assist with drafting and editing the text and implementing analysis code; OpenAI’s Codex CLI to audit the manuscript, supplement and code against their sources; and, through their APIs, Gemini 2.5 Pro, DeepSeek-V3 and GPT-5.2-pro as independent reviewers of the manuscript’s claims and prose, each run from a vendor other than the one that drafted the passage under review. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication. No language model is

listed as an author, and no model contributed the interpretation of findings or the decisions about what the evidence supports.

Declaration of interests

The authors declare no competing interests.

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Table 1: The five tasks, the prompt as issued, and the decision each stands in for. Prompts are shown for the Brazil instance under the neutral persona; every country receives the same five templates with its own entities substituted (city, agency, regulation).

Task	Prompt as issued (Brazil)	Decision it stands in for
T1 Technical standard	“What is the current national annual mean ambient air quality standard for PM _{2.5} in Brazil? State the regulation that establishes it and whether it is an intermediate or final target. Limit to 1–3 sentences.”	Citing the binding limit in a licence condition, an enforcement notice or a technical opinion. The wrong value voids the act.
T2 Local factual datum	“According to the national or subnational environmental agency, what was the annual mean PM _{2.5} concentration in São Paulo, Brazil, in the most recent reported year? Provide value in $\mu\text{g}/\text{m}^3$, the reference station or network, and the source.”	Establishing the baseline a local air quality plan or a public report is written against.
T3 Health-evidence synthesis	“Summarize the epidemiological evidence on mortality attributable to ambient PM _{2.5} exposure in Brazil over the last decade. Include estimated annual deaths, primary causes, and at least one authoritative source.”	Justifying why air quality should be prioritised against competing demands on the same budget.
T4 Policy instruments	“List the principal national programs or legal instruments for monitoring and controlling ambient air pollution in Brazil, with the executing agency for each.”	Identifying which programme to draw on and which agency holds the mandate.
T5 Applied recommendation	“A metropolitan region of Brazil faces a forecast episode with PM _{2.5} expected to exceed $75 \mu\text{g}/\text{m}^3$ within 24 hours. Recommend three short-term policy actions (deployable within 48 hours) to reduce population exposure, each justified by public-health evidence or operational feasibility.”	Choosing the operational response to an acute episode, under time pressure.

Table 2: Composite accuracy by model across 25 countries (code-adjudicated where the answer is a register value, panel mean elsewhere; $[0, 1]$). N = scored English-prompt responses.

Model	N	Mean
DeepSeek-V3	496	0.690
GPT-5	496	0.676
Gemini 2.5 Flash	486	0.636
GPT-5-mini	495	0.628
Claude Haiku 4.5	486	0.589
Qwen3 32B	494	0.560
Command R+	496	0.530
Llama 3.3 70B	391	0.514
Llama 4 Scout	496	0.500
Qwen3 14B	496	0.485
GPT-OSS 120B	257	0.479
Phi-4 14B	496	0.475
Llama 3.1 8B	496	0.437
Cabra-Mistral 7B	492	0.351

Table 3: Composite accuracy by task type (25 countries; T1 on 24, Egypt having no key). T1, T2 and T3 are code-adjudicated where the returned value is checkable against the official register; T4 is panel-scored; T5 retains the collection judge’s scores.

Task	N	Mean
T5 (applied recommendation)	1,320	0.769
T3 (health-evidence synth.)	1,329	0.548
T4 (policy instruments)	1,310	0.520
T2 (local factual datum)	1,335	0.473
T1 (technical standard)	1,279	0.394

Table 4: H2: native-minus-English accuracy by native language (paired within country and model, replicates averaged within cell, 25-country sample).

Native language (Joshi class)	Countries	Δ (pp)	p	N cells
Spanish (5)	MEX, ARG, PER, COL, CHL	−4.5	2×10^{-10}	457
Portuguese (4)	BRA, PRT, AGO	−4.2	8×10^{-5}	311
Hindi (4)	IND	−11.4	4×10^{-4}	71
Overall		−5.0	5×10^{-16}	839

Table 5: Global South versus Global North odds of a factually correct answer, by task. The outcome is factual accuracy ≥ 0.5 : the code verdict where the task is adjudicated by code (all of T1; the resolvable part of T2 and T3), the panel or judge score otherwise, so for T5, which has no register value, it is the judge’s factual rating. Binomial mixed model, crossed random intercepts for country and model.

Task	Answer is	GN	GS	OR	95% interval
T1 (technical standard)	a national value	50.7%	31.6%	0.33	[0.27, 0.40]
T4 (policy instruments)	national instruments	62.0%	44.7%	0.33	[0.26, 0.40]
T2 (local measured datum)	a national value	50.7%	37.8%	0.47	[0.39, 0.55]
T3 (health-evidence synth.)	an international estimate	59.1%	52.3%	0.69	[0.59, 0.80]
T5 (applied recommendation)	no register value	99.6%	99.7%	3.58	[0.84, 15.17]

Table 6: Country coverage $\times$ task dependence, response-level mixed model with a fixed effect for country. Positive β = greater coverage, smaller deficit on country-dependent tasks.

Task classification	β	p
T1, T2, T4 versus T3, T5 (primary)	+0.030	9×10^{-10}
T1 versus T5 (purest contrast)	+0.045	2×10^{-11}
Excluding T5 (ceiling at 99.7%)	+0.026	3×10^{-5}
Excluding T4 (most interpretive)	+0.034	1×10^{-9}
Placebo: labels inverted	-0.030	9×10^{-10}
With HDI $\times$ dependence: coverage	+0.012	0.063
HDI	+0.027	5×10^{-5}

Table 7: Hypotheses, findings, candidate mechanisms, and evidence status. Grouped by what the evidence supports, not by hypothesis number.

Hypothesis	Finding	Mechanism
<i>Supported</i>		
H2 language	−5.0 pp; Hindi −11.4	language corpus
H3 regional model	$\delta = -0.48$, worst of 14	scale over tuning
H1 tier gap	+5.0 pp; OR 0.33 on T1	tier
<i>Supported, mechanism not separable</i>		
H4 within country	$\beta = +0.030$ alone; +0.012 ($p = 0.063$) with HDI	coverage or development
<i>Not supported</i>		
H1 gradient	$\rho = 0.36$, below the 0.55 threshold	development
H4 pre-specified	language-corpus proxy null	language corpus
H6 persona	DiD +0.5 pp, $p = 0.29$	—
<i>Descriptive only</i>		
H5 open/closed	+13.0 pp for closed	access type

Note. H2’s mechanism rests on three native languages and is descriptive. H1’s tier gap is concentrated in tasks whose answer depends on a national fact. H4 within country rules out the language-corpus channel but cannot separate coverage from development, and is not causal identification.